Vol. 00, e000000 (20XX)
doi: 10.2142/biophysico.bppb-v00.0000

Biophysics and Physicobiology

https://www.jstage.jst.go.jp/browse/biophysico/

Regular Articles

Transfer learning with simulated variants and calculated quantities for nanobody melting-temperature prediction

Taihei Murakami[2,3*], Kentaro Sasaki[2*], Soichiro Oda[2], Kazuma Okada[2], Yasuhiro Matsunaga[1,2]

[1] *RIKEN Center for Computational Science, Kobe, Hyogo 650–0047, Japan*
[2] *Saitama University, Saitama, Saitama 338–8570, Japan*
[3] *Epsilon Molecular Engineering, Inc., Saitama, Saitama 338–8570, Japan*

Received XX Xxx, 20XX; Accepted XX Xxx, 20XX;
Released online in J-STAGE as advance publication XX Xxx, 20XX
Edited by Xxx Xxx

Experimental nanobody melting-temperature (Tm) data are limited. Computed stability values can be used during model training, but they describe quantities related to Tm rather than Tm itself. We asked how the choice of simulated variants and calculated quantity affects their usefulness. We trained multi-task models that used a shared ESM2 sequence model for experimental Tm and one computed label at a time. We used 57 sequences for training, 114 for validation, and 396 for held-out testing. Each label and encoder setting was tuned separately using only the Tm validation set, then compared on the same test proteins. The choice of simulated variants mattered. An MD native-contact label reduced frozen-encoder Tm error by 0.20°C when computed for local mutation scans on two fixed structures, compared with 0.05°C for a heterogeneous structure panel. The measured quantity also mattered. Free-energy perturbation (FEP) labels improved both frozen and fine-tuned encoders, reducing the fine-tuned mean absolute error from 6.55°C to 6.40°C. Native-contact labels helped only the frozen encoder, while Rosetta and ThermoMPNN scores gave little or no improvement. These results show that simulation data for experimental prediction should be built from carefully chosen variants and from quantities that are closely related to the experimental property.

Key words: nanobody thermostability, melting-temperature prediction, multi-task transfer learning, protein language model, molecular simulation labels

◄ Significance ►

We tested whether computed stability values improve nanobody melting-temperature prediction when only a small experimental training set is available. Native-contact values from local mutation scans gave a larger gain than values from a heterogeneous structure panel. FEP mutation free energies improved models with both frozen and fine-tuned sequence encoders, whereas native-contact values helped only the frozen model. These results show that both the simulated variants and the calculated quantity need to be considered when preparing training data.

Introduction

Machine-learning models for molecular design need measured examples, but many experiments are slow or expensive. Properties calculated with quantum chemistry, molecular simulation, or other physical models can be used as additional training signals. This approach is often called simulation-to-real (Sim2Real) learning. It has been used in materials sci-

*These authors contributed equally.
Corresponding author: Yasuhiro Matsunaga, RIKEN Center for Computational Science, Kobe, Hyogo 650–0047, Japan. ORCID iD: https://orcid.org/0000-0003-2872-3908, e-mail: ymatsunaga@riken.jp

ence to combine large computed data sets with smaller experimental data sets [1–3]. In some cases, prediction improves as more computed data are added [3]. This outcome is not guaranteed. A calculation may use an imperfect model, sample the wrong systems, or measure a property that is only weakly related to the experiment.

Protein stability makes this problem clear because several common labels have different physical meanings. Melting temperature (Tm) is the experimental temperature at which folded and unfolded populations are balanced under a given assay. A mutation free-energy change ($\Delta\Delta G$) instead describes the change in folding free energy caused by a sequence change. Rosetta and learned stability models provide scores on their own scales, while molecular dynamics (MD) can provide structural measurements such as the fraction of native contacts that remain during a trajectory. All of these labels concern protein stability, but none is a Tm measurement. A poorly matched computed label may therefore leave Tm prediction unchanged or make it worse.

We study this question in nanobodies, single-domain VHH antibodies used as small and modular binding proteins [4,5]. Thermal stability affects their expression, storage, and formulation [6–8]. Direct measurements and detailed simulations of stability remain costly [9–12]. Protein language models such as ESM2 can learn useful sequence features from large protein databases [13–15], and several methods use these features to predict Tm [16–21]. However, a sequence model does not remove the need for measured Tm values from the protein family of interest.

Large protein-stability data sets are becoming available. For example, Tsuboyama et al. measured folding stability for about 776,000 natural and designed protein domains by cDNA-display proteolysis [22]. Those domains are 40–72 residues long and differ from VHH domains in both sequence and measurement type. Models trained on small domains also transfer less well to longer proteins [23]. Nanobody-specific data and validation are therefore still needed [24–26].

The open question is not simply whether computed labels can be added to a Tm model. It is which variants should be simulated and which quantity should be calculated. A mixed set of unrelated sequences may contain unwanted differences such as sequence length. A mutation scan on one or two fixed structures gives a more controlled set of sequence changes. Even with the same variants, a mutation free energy, a Rosetta score, and an MD native-contact value may differ in how well they relate to Tm. These choices are made before model training, so they should be tested as part of the simulation plan.

Here, we compare several computed labels in the same low-data Tm task. The labels are mutation free energies from free-energy perturbation (FEP), Rosetta mutation scores, ThermoMPNN scores, Rosetta scores for ESM2-proposed or random variants, and MD native-contact values. For the MD label, we compare a mixed sequence set with local mutation scans on the same 1MEL and 4IDL structures used for FEP. Each computed label is trained as a second task that shares an ESM2 sequence model with Tm prediction (Fig. 1). We tune every label separately using only the experimental Tm validation set. We then compare the selected models on the same held-out Tm test proteins using paired errors. In this paper, "transfer" means that adding a computed label lowers the held-out Tm error.

Two results stand out. First, the choice of simulated variants matters. The MD native-contact label gives a much larger improvement with a frozen encoder when it is calculated for local mutation scans on fixed structures than for a mixed set of sequences. Second, the calculated quantity matters. FEP labels improve Tm prediction with both frozen and fine-tuned encoders, whereas native-contact labels help only the frozen encoder. Rosetta and ThermoMPNN scores give little or no improvement after separate tuning. These results show that useful simulation data require two choices before calculation: a controlled set of sequence changes and a quantity that is closely related to the experimental property.

Materials and methods

Study design

We asked whether computed protein-stability labels could improve nanobody melting-temperature (Tm) prediction when only a small experimental training set was available. Experimental Tm was always the task used to choose a model. Each computed label was added as a second regression task that shared all or part of the sequence model with the Tm task. We compared each label with a Tm-only model trained and selected in the same encoder setting.

All choices were made with the experimental Tm validation set. The test set was used only after the model form and training settings had been fixed. Errors on the computed labels were not used for checkpoint or hyperparameter selection.

Experimental Tm data

We used the thermo-seq task from NbBench, a public nanobody benchmark [25]. We reassigned its three published partitions before the comparisons in this study so that the smallest partition was used for training. The resulting split contained 57 training, 114 validation, and 396 test sequences. This split models the setting that motivated the study: few Tm measurements for fitting a model, but enough held-out proteins for paired comparisons. The training set fit model

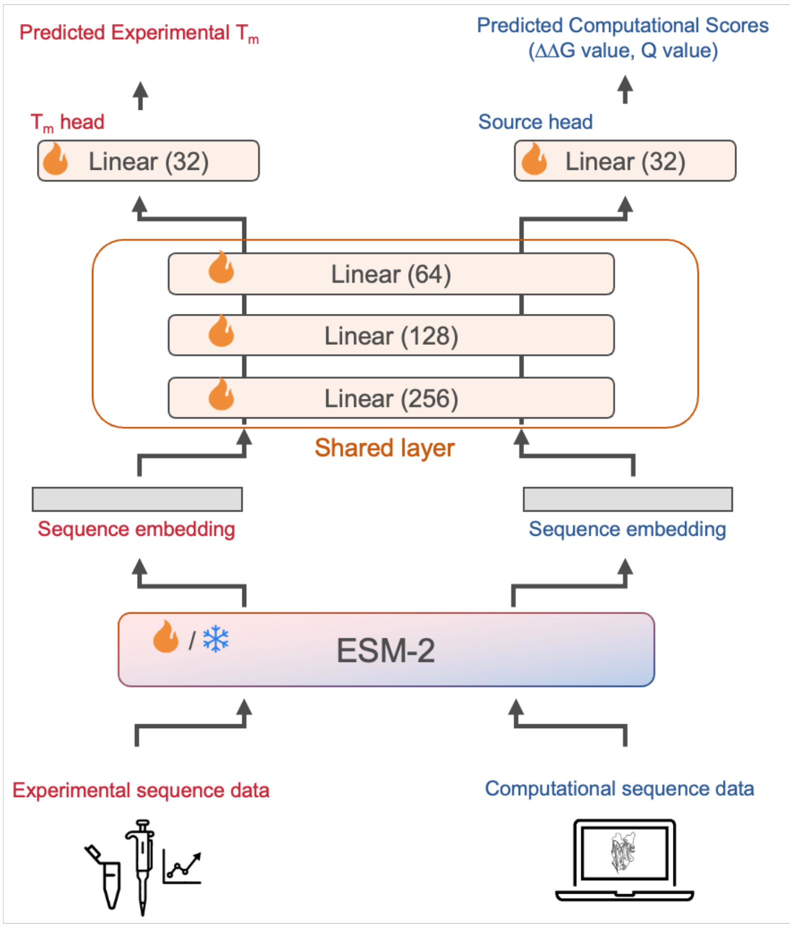

Figure 1 Multi-task model used to test computed stability labels. Experimental sequences and sequences with computed labels pass through the same ESM2 model and shared multilayer perceptron. Separate scalar heads predict Tm and the computed label. The ESM2 model is either kept frozen or fine-tuned. Models are selected using experimental Tm validation data, and the final comparison uses held-out Tm test data.

parameters, the validation set selected checkpoints and hyperparameters, and the test set was kept for final evaluation.

Tm values were stored in degrees Celsius. A pipeline consisting of a robust scaler followed by min–max scaling to [0,1] was fit on the 57 training values only. It was then applied to the training and validation labels. Predictions were returned to degrees Celsius before validation and test errors were calculated. Validation and test Tm values therefore did not affect the scaling step.

Computed-label tables and sampling

We tested seven sources of computed labels: FEP mutation free energies, matched MD native-contact values, Rosetta mutation scores, ThermoMPNN predictions, Rosetta scores for random variants, Rosetta scores for ESM2-proposed variants, and MD native-contact values for a heterogeneous structure panel. These values were treated as related measurements, not as estimates of Tm.

FEP, Rosetta, and ThermoMPNN used two single-mutation tables based on the 1MEL and 4IDL structures. Each contained 435 and 409 rows, respectively. The matched MD tables used the same two structures and mutation scans and contained 431 1MEL rows and 406 4IDL rows that reached the common analysis time. The random and ESM2-proposed variant sets each contained 1000 two-mutation sequences per structure. The heterogeneous MD table contained 1143 structure-derived rows.

For all two-table sources, the notation n means a cap of n rows from *each* structure table, not n rows in total. Thus, $n = 320$ gives at most 640 computed-label rows. Rows were sampled independently from the 1MEL and 4IDL tables with the run seed. The diverse MD source had one table, so n is the total number sampled from that table. After sampling, each computed-label table was divided at random into 80% training and 20% monitoring rows. These monitoring rows were not passed to the trainer for checkpoint selection; only the experimental Tm validation rows were used for that purpose. The relation between sequence length and the diverse-panel MD label is shown in Supplementary Fig. S1; exact sequence overlaps are reported below.

The 1MEL and 4IDL tables were kept as two tasks because their score ranges can differ. Models used either two separate output heads or one output head with a learned structure indicator, as described below. Every processed table supplied an amino-acid sequence and one scalar label scaled to [0,1]. FEP, Rosetta, and ThermoMPNN scores were sign-reversed so that larger values indicated a more favorable change, then processed separately for each structure with robust scaling, a Yeo–Johnson transform, and min–max scaling. MD contact values were min–max scaled with larger values indicating greater contact retention.

We also checked exact sequence matches between every computed-label table and the NbBench splits. There were no exact matches for the FEP, matched MD, Rosetta, ThermoMPNN, random-variant, or ESM2-proposed tables. The diverse MD table contained 2 training, 4 validation, and 8 test sequences also present in NbBench. We retained these rows and report the overlap because it limits the interpretation of that comparison.

FEP mutation free energies

The FEP labels were mutation free-energy changes, $\Delta\Delta G$, for the 1MEL and 4IDL mutation scans. Positive raw $\Delta\Delta G$ denotes a destabilizing mutation. The learning tables therefore used the sign-reversed value before the scaling described above.

The fixed FEP tables were produced with alchemical simulations in NAMD using the CHARMM36 protein force field and explicit TIP3P water [27–29]. The starting coordinates were the 1MEL and 4IDL crystal structures. Mutation systems were prepared with the VMD AlaScan workflow and CHARMM hybrid-residue topologies [30, 31]. Representative archived input files show a 2-fs timestep, particle mesh Ewald electrostatics, bonds to hydrogen held fixed, a temperature of 300 K, and a pressure of 1 atm [32, 33].

Each forward and backward transformation covered 20 windows of width $\Delta\lambda = 0.05$ between 0 and 1. A window contained 500,000 steps (1 ns), with its first 100,000 steps assigned to equilibration. The folded-state free energy was combined with a residue-specific unfolded-state reference to obtain $\Delta\Delta G$ relative to the wild type. The present learning pipeline starts from the processed CSV files and does not rerun the FEP calculations.

MD native-contact labels

Both MD labels were based on the smooth native-contact fraction Q [34]. For a native contact with reference distance d_{ij}^0 and distance $d_{ij}(t)$ at time t, its contribution was

$$q_{ij}(t) = \left[1 + \exp\left\{\beta[d_{ij}(t) - \lambda d_{ij}^0]\right\}\right]^{-1},$$

with $\beta = 50\ \mathrm{nm}^{-1}$ and $\lambda = 1.8$. Native contacts were pairs within 0.45 nm in the reference frame and separated by more than three residues. Q was the mean over the selected contacts and frames.

Matched mutation scans. We simulated the 1MEL and 4IDL wild types and the same single-mutant sequences used in the FEP comparison. Each protein was placed in a cubic box of OPC water with a 15-Å buffer and 0.15 M salt, and was described by Amber ff19SB with hydrogen-mass repartitioning [35, 36]. OpenMM was used for minimization, equilibration, and production [37]. The recorded run settings were 0.2 ns NVT followed by 0.8 ns NPT at 300 K, 0.5 ns restrained NVT at 400 K, and restraint-free NVT production at 400 K. All stages used a 4-fs timestep; production coordinates were written every 10 ps. Although some runs continued to 100 ns, the label used the first 4000 production frames (40 ns) for every variant.

For this source, the reference distances came from production frame 0 and all backbone atoms selected by MDAnalysis were eligible for contacts. We computed $\Delta Q = Q_{\mathrm{mutant}} - Q_{\mathrm{WT}}$ within each structure and then min–max scaled it. Subtracting one constant wild-type value does not change a min–max-scaled label, so scaled Q and scaled ΔQ are identical within each structure table.

Diverse nanobody panel. The comparison panel was built from SAbDab structures [38]. One verified heavy-chain nanobody was taken per PDB entry when an IMGT-renumbered structure was available. Preparation used PDBFixer and pH-aware protonation with pdb2pqr/PROPKA when available [39, 40]. The systems used ff19SB, OPC water, 0.15 M salt, and OpenMM. After 300 K NVT and NPT equilibration, two restrained 400 K NVT stages preceded 100 ns of unrestrained

400 K NVT production. Requiring a usable topology and trajectory, a sequence of at least 50 residues, and at least one selected contact gave 1143 rows.

The diverse-panel label used production frame 0 as its reference and averaged approximately the last 30 ns. Its contact set contained backbone heavy-atom pairs for which at least one residue was Asp, Glu, Gln, Asn, Arg, Lys, or His. It was min–max scaled across the single table. The matched and diverse studies used similar simulation time per sequence and therefore compare two ways of spending a similar computational budget. They used the same Best–Hummer form and the same temperature, although their contact selection and averaging window were not identical. We therefore treat this as a comparison of complete simulation plans, rather than as a one-variable test of sequence choice alone.

Rosetta, ThermoMPNN, and variant-selection comparisons

Rosetta scores were calculated with the `ddg_monomer` application and the `ref2015` score function [41,42]. The archived command used 50 iterations, Cartesian minimization, minimum-score aggregation, and full-atom input. The same settings were used for the 1MEL/4IDL mutation tables and for the two sets of two-mutation variants.

For the random set, mutation positions and replacement amino acids were sampled uniformly, excluding cysteine as a replacement, until 1000 unique two-mutation variants had been obtained for each structure. For the ESM2-guided set, the 650M-parameter ESM2 model generated 100,000 two-mutation variants per structure [14]. Variants were ranked by pseudo-log-likelihood, and the top 1% were scored with Rosetta. ThermoMPNN predictions were made for the same 435 and 409 single-mutation sequences used by the structure-matched tables [43]. The current loader reads the two processed ThermoMPNN tables, rather than the larger raw Colab outputs stored with them.

Sequence model and regression heads

The main encoder was the 8M-parameter ESM2 model `facebook/esm2_t6_8M_UR50D` [14]. Sequences were padded or truncated to 160 residues. The first-token representation was used as the sequence vector. In the shared model, this vector passed through linear layers of size 256, 128, and 32, with ReLU activation and dropout after the first two layers. The Tm head and each computed-label head were separate 32-to-1 linear layers.

The model search also included residual and latent forms. Both gave Tm and computed-label examples separate 256–128–32 paths. The residual form added a sigmoid-gated correction to a base Tm prediction. The latent form predicted a Tm correction from the Tm vector, the second path's vector, and their elementwise product. For two-table sources, we compared separate 1MEL and 4IDL heads with a single head conditioned on a learned structure embedding. In the residual and latent implementations, the computed-label loss received a detached encoder vector by default, whereas the Tm path could still update the encoder. In the shared form, losses from all tasks could update the encoder when it was not frozen.

We tested two encoder settings. In the frozen setting, all ESM2 parameters were fixed and sequence vectors were computed once before head training. In the hot setting, all ESM2 parameters could change. AdamW used a smaller learning rate for ESM2 than for the newly initialized layers. Selected controls used 35M- and 650M-parameter ESM2 encoders.

Loss and model selection

Each row had one task identifier and contributed loss only through that task's output head. We used Huber loss with $\delta = 1$ on the scaled labels. By default, the losses were combined with learned uncertainty weights [44]. For task k, loss L_k, and learned log-scale s_k, the term added to the total loss was

$$\frac{L_k}{2\exp(2s_k)} + s_k.$$

Fixed task weights were tested as controls.

Models were trained with AdamW, cosine learning-rate decay, a batch size of 16, 100 warmup steps, and at most 400 epochs. Mixed-precision training was used on CUDA devices. A checkpoint was saved and evaluated once per epoch. The checkpoint with the lowest mean squared error on the 114 experimental Tm validation rows was retained, and training stopped after 15 epochs without improvement.

Every label source was tuned separately in the frozen and hot settings. The search had two stages. Stage 1 crossed three model forms (shared, residual, and latent) with two head forms (separate and structure-conditioned); Tm-only models searched the three model forms without a computed-label head. Stage 2 held the chosen form fixed. For hot models it tested encoder learning rates of 1×10^{-4}, 5×10^{-5}, 3×10^{-5}, and 1×10^{-5}. For frozen models it tested head learning rates of 1×10^{-3}, 5×10^{-4}, and 2×10^{-4}. Both searches crossed dropout values of 0.05, 0.15, and 0.30 with weight decay values of 0.02 and 0.10.

Each candidate was ranked by the MAE of a three-seed ensemble on the experimental Tm validation set, using $n = 320$ rows per structure table. The selected setting was then trained with five seeds for the final test result. Label-count curves used the same selected setting at $n = 20$, 80, 160, and 320 per structure; they were not retuned at each point.

Test evaluation and statistics

For each condition, the five models predicted all 396 test sequences. Their predictions were averaged in scaled space and then returned to degrees Celsius. The main measure was mean absolute error (MAE). We also calculated root mean squared error, R^2, and Spearman and Pearson correlations.

Uncertainty was estimated by nonparametric bootstrap over test proteins. For one condition, we resampled its 396 absolute errors with replacement 10,000 times, using random seed 42, and calculated MAE for each sample. The reported 90% confidence interval is the 5th to 95th percentile range.

For a comparison, the same resampled test indices were applied to the candidate and reference error vectors. We calculated ΔMAE as candidate MAE minus reference MAE for every sample. Negative values therefore favor the candidate. Main figures report the mean paired ΔMAE and its 90% confidence interval. We do not use a separate p-value threshold for the main claims.

Software and saved outputs

The data loader, training code, model settings, and ensemble evaluation are in `prepare.py`, `train.py`, and the scripts distributed with the repository. Each final run stores its command-line settings, predictions, absolute errors, summary measures, and bootstrap results in machine-readable files. The figure scripts read those saved files directly. The released workflow begins with the fixed processed CSV tables; it does not repeat the raw FEP, Rosetta, ThermoMPNN, or MD calculations.

Results and discussion

Computed labels were judged only by experimental Tm

We first set up a common test for all computed labels (Fig. 1). Experimental Tm was the prediction target. FEP, MD, Rosetta, and ThermoMPNN values were separate regression tasks; they were not treated as extra Tm measurements. Each task used the same ESM2 sequence model and either shared or task-specific regression layers. We tested a frozen ESM2 encoder and a fully fine-tuned, or hot, encoder.

Only experimental Tm was used to choose a model. The 57 training proteins fit the model, the 114 validation proteins selected the architecture and training settings, and the 396 test proteins were used after selection. Every computed label was tuned separately in the frozen and hot settings. We then compared absolute errors on the same test proteins. This was necessary because a model can predict a computed value well without improving Tm.

The selected Tm-only model had a test MAE of 7.23°C with a frozen encoder and 6.55°C with a hot encoder. The lower error of the hot baseline shows that fine-tuning ESM2 was useful even with only 57 measured training proteins. Against these two baselines, we asked two questions. First, does the set of sequences chosen for simulation affect transfer? Second, does the computed quantity affect whether the benefit remains when ESM2 is fine-tuned?

Local mutation scans gave a stronger MD result

We compared two choices made before running MD. The heterogeneous structure panel spanned single-domain antibodies of different lengths. The matched scan used local mutations of the fixed 1MEL and 4IDL structures. Both calculations used 400 K MD, similar simulation time per sequence, and labels from the Best–Hummer native-contact Q family. Thus, the comparison asks how a similar amount of simulation should be distributed over sequence space.

The heterogeneous and matched studies were run and tuned as separate series, so their absolute MAEs cannot be compared directly. Figure 2a instead shows the paired change from the Tm-only model in each series. In the heterogeneous series, the MD label changed MAE by −0.049°C with a frozen encoder (90% CI, −0.085 to −0.012°C) and by +0.116°C with a hot encoder (90% CI, +0.012 to +0.224°C). The frozen gain was small, and the hot model became worse. In the matched series, the frozen change was −0.195°C (90% CI, −0.336 to −0.055°C), a larger observed gain than in the heterogeneous series. The hot change was +0.029°C, with a confidence interval that included zero.

Several differences between the two sequence sets may contribute to this result. The heterogeneous set ranged from 58 to 461 residues, and its Q value had a Pearson correlation of $r = +0.13$ with sequence length. Length could therefore provide an unwanted shortcut (Supplementary Fig. S1). The matched scans kept length fixed within each structure and supplied local directions: one mutation and its change in contact retention. This is a simpler learning problem. However,

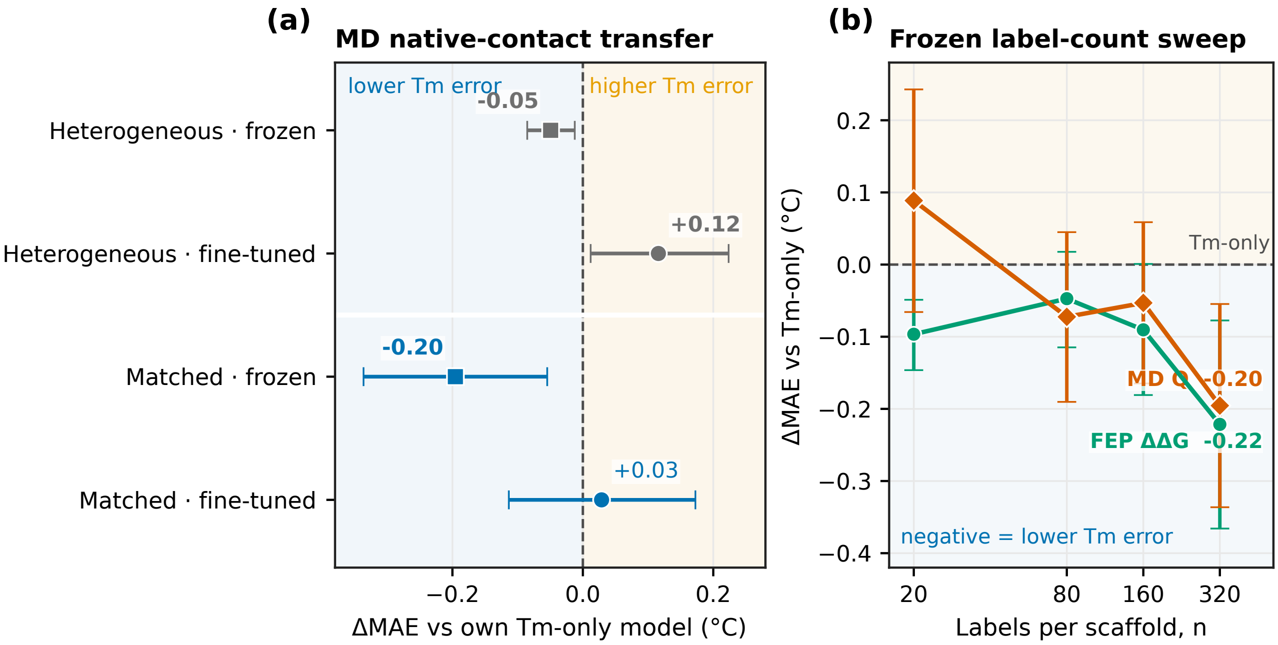

Figure 2 Matched mutation scans show a larger frozen-encoder gain from native-contact Q. The heterogeneous plan distributed simulations across variable-length structures, whereas the matched plan used local mutations of 1MEL and 4IDL. Both used 400 K MD, similar simulation time per sequence, and labels from the same Q family; contact selection and averaging windows differed. (a) Paired MAE change for each simulation plan and encoder setting. Zero means equal MAE to that series' own Tm-only model, and negative values mean lower Tm error. Because the two series were tuned independently, their absolute MAEs are not compared. The heterogeneous and matched series use fixed 10-seed and five-seed ensembles, respectively. Bold values have nominal 90% paired bootstrap intervals that exclude zero; intervals are not adjusted for multiple comparisons. (b) Paired MAE change from the frozen Tm-only model over four tested label counts, shown on a $\log_2$ count axis. Here n is the number sampled from each structure table. Points and whiskers show the paired change and its nominal 90% bootstrap interval over the 396 test proteins.

the present data do not show that length alone caused the difference. The two MD pipelines also differed slightly in contact selection and in the trajectory frames used for averaging, as detailed in Materials and Methods. The result therefore compares the two complete simulation plans, not a single changed variable.

Wild-type referencing does not explain the matched result. Its label was $\Delta Q = Q_{\mathrm{mutant}} - Q_{\mathrm{WT}}$, but the same wild-type constant was subtracted from every mutation in a structure table. The following min–max scaling removes this constant, so scaled Q and scaled ΔQ are identical within that table.

The frozen label-count curves give a second view of the matched scan (Fig. 2b). Each point is the paired MAE change from the same frozen Tm-only model; zero therefore means equal test MAE. Here n is the number of labels sampled from each structure table. At $n = 20$, the changes were −0.10°C for FEP and +0.09°C for matched MD. At $n = 320$, both sources were better than Tm-only: their paired changes were −0.22°C for FEP and −0.20°C for MD. The direct FEP-minus-MD difference was only −0.03°C (90% CI, −0.21 to +0.16°C). The four tested settings were not monotonic enough to support a scaling law. With a frozen encoder, the matched MD scan performed as well as FEP at the largest setting.

The main conclusion from Fig. 2 is narrow. The local mutation scans on fixed structures showed a larger frozen MD gain than the heterogeneous panel. They did not make the MD label helpful in the fine-tuned model. Sequence choice matters, but it is not the only choice that matters.

Only FEP helped both frozen and hot models

We next compared all computed labels after separate tuning (Fig. 3a). With a frozen encoder, FEP and matched MD gave the clearest gains. FEP reached 7.01°C MAE, a paired change of −0.221°C from Tm-only (90% CI, −0.366 to −0.077°C). Matched MD reached 7.03°C, a change of −0.195°C. ThermoMPNN was third at 7.09°C, but its change of −0.141°C had a confidence interval that included zero. Plain Rosetta, random variants scored by Rosetta, and ESM2-proposed variants scored by Rosetta were close to or worse than the Tm-only model.

The order changed with a hot encoder. FEP reached 6.40°C, the lowest MAE in the study. Its paired change from the 6.55°C Tm-only model was −0.153°C (90% CI, −0.295 to −0.009°C). No other computed label improved the hot baseline. Matched MD, ThermoMPNN, plain Rosetta, and random variants plus Rosetta had changes from +0.029 to +0.144°C, all with intervals that included zero. ESM2-proposed variants plus Rosetta were worse by +0.411°C (90% CI, +0.181 to +0.651°C).

Figure 3b directly compares the two all-atom labels. The plotted quantity is MAE(FEP) minus MAE(matched MD), so zero means equal test MAE and negative values favor FEP. The frozen difference was −0.026°C (90% CI, −0.211 to +0.159°C), which did not separate the labels. With a fine-tuned encoder, FEP was better by −0.182°C (90% CI, −0.359 to −0.009°C).

The same separation appears across label counts (Fig. 3c), where each curve is a paired change from the fine-tuned Tm-only model. At $n = 20$, matched MD was worse than Tm-only by +0.47°C (90% CI, +0.29 to +0.65°C). Its error decreased as n increased, but at $n = 320$ its paired change was still +0.03°C and its interval included zero. FEP changed from +0.02°C at $n = 20$ to −0.15°C at $n = 320$. These four tested settings show recovery and crossover; they do not establish a power law.

FEP $\Delta\Delta G$ and Tm are different measurements. $\Delta\Delta G$ gives the change in folding free energy caused by a mutation, whereas Tm is the temperature at which folded and unfolded states are balanced under an experimental condition. They are nevertheless linked by protein folding thermodynamics. In this data set, the measured Tm examples supply absolute temperature values across nanobodies, while the FEP scans supply relative information about which local sequence changes stabilize or destabilize two structures. The matched MD scan also supplies local information, but contact retention is a more indirect measure of folding stability. This difference is a plausible reason why the two labels tie in the frozen model but separate in the hot model.

We describe this pattern as a difference in transfer depth. Here, “deeper” means only that a benefit remains when the pretrained encoder is allowed to change. A recent preprint from our group gives a more concrete basis for this interpretation [45]. That study used sparse autoencoders to examine ESM2 after fine-tuning on nanobody Tm. It found interpretable features linked to surface charge, hydrophobic-core packing, the VHH tetrad, and disulfide bonds. Thus, fine-tuning on nanobody Tm can produce a representation with specific, physically meaningful stability features rather than only a better regression fit.

Together with that result, the present frozen–hot difference suggests a testable hypothesis: mutation free energies may support the same kinds of local stability features that a Tm-fine-tuned encoder learns, while a single contact-retention value may provide a weaker guide for updating them. This remains an interpretation, not a result measured here. We did not apply sparse autoencoders or compare hidden states in the present models, and the selected FEP and MD models did not always use the same regression architecture. A direct next test is therefore to compare the sparse features learned by Tm-only, Tm+FEP, and Tm+MD hot encoders and ask which stability features appear, disappear, or change strength.

Controls, scope, and next tests

Several controls limit alternative explanations. Neither the 35M nor the 650M Tm-only model reached the absolute MAE of the 8M model trained with FEP labels, so encoder size alone did not substitute for the computed label. Within each larger-size exploratory control, however, adding FEP labels also reduced MAE relative to its corresponding Tm-only model (Supplementary Fig. S2a). An analytical correction for periodic-image effects in charge-changing FEP mutations had almost no effect on the scaled labels (Supplementary Fig. S2b). Tests that clipped the largest FEP values or added the available Phe scan did not improve transfer. These checks support the use of the original FEP tables, although they do not prove that every large $\Delta\Delta G$ value is quantitatively accurate.

The other label sources provide an important negative result. ThermoMPNN gave at most a weak frozen gain, and none of the Rosetta sources improved both encoder settings. ESM2-proposed variants scored by Rosetta did not beat random variants scored by the same method. We therefore treat these sets as comparisons of sequence choice and score type. The results do not support a claim that the ESM2 proposals form a successful protein-design loop.

The findings agree with the main idea of Sim2Real learning: computed data are useful when they share information with the measured task [3]. The protein case adds an earlier decision. Before a label can transfer, the simulated sequences must make the desired relationship easy to learn. A large mixed set is not automatically better than a smaller local scan. After the sequences are chosen, the computed quantity still matters. FEP and matched MD used roughly similar simulation time per variant here, but we did not record wall-clock cost in a form that supports a general cost claim.

This study has several limits. The Tm training set contains only 57 nanobodies, and the matched labels cover only two structures. The heterogeneous and matched MD pipelines differ in more than their sequence sets, and the heterogeneous

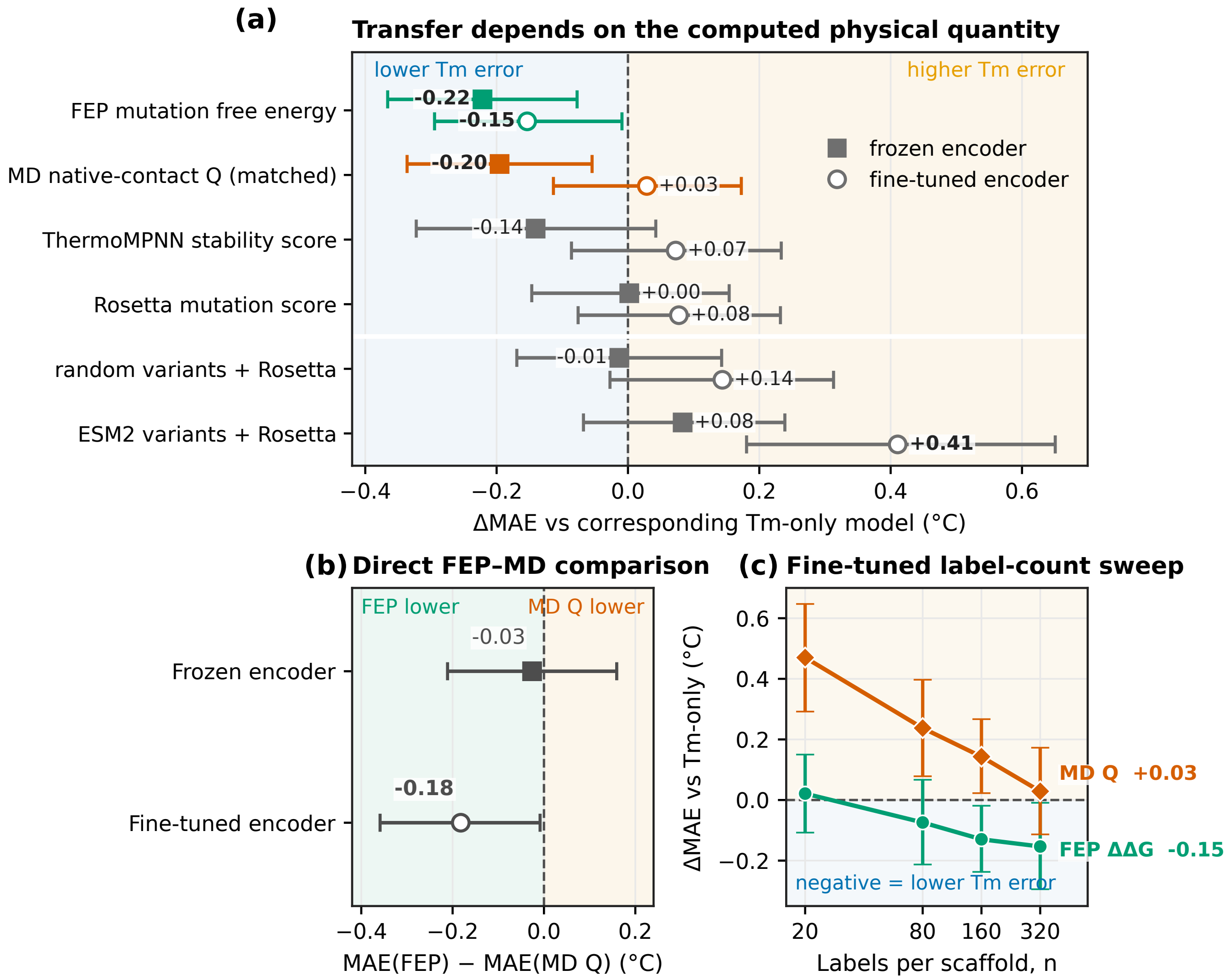

Figure 3 FEP is the only computed label that improves both encoder settings. (a) Points show the paired MAE change from the corresponding Tm-only model for every tuned label. Negative values mean lower Tm error. Filled squares and open circles denote frozen and fine-tuned encoders, respectively; whiskers are nominal 90% paired bootstrap intervals. Bold values have intervals that exclude zero; intervals are not adjusted for multiple comparisons. (b) Direct FEP-minus-MD comparison. Zero means equal held-out Tm MAE, negative values mean lower error for FEP, and positive values mean lower error for MD native-contact Q. The bold value has a nominal 90% paired interval that excludes zero. (c) Paired MAE change from the fine-tuned Tm-only model over four tested label counts on a $\log_2$ count axis. FEP moves below zero, whereas matched MD approaches zero from higher error. Here n is the number sampled from each structure table. Whiskers are nominal 90% paired bootstrap intervals over the 396 test proteins.

table includes eight exact matches to the test set. The source list does not cover every useful simulation measurement. The confidence intervals also show that small differences among the middle-ranked labels remain uncertain. Finally, we did not make new Tm measurements or test predictions prospectively.

The next test should extend matched mutation scans to more nanobody structures while keeping the simulation and label-extraction settings fixed. It should compare several physical quantities on exactly the same variants and then test the selected Tm model on new experimental measurements. Such a study would separate sequence choice from label choice more cleanly and show whether the FEP result extends beyond 1MEL and 4IDL.

Conclusion

Computed labels improved low-data nanobody Tm prediction only in specific settings. The sequences chosen before simulation mattered: MD native-contact labels from local mutation scans on 1MEL and 4IDL gave a clearer frozen-encoder gain than labels from a heterogeneous sequence screen. The calculated quantity also mattered. FEP mutation free energies improved both frozen and fine-tuned models, whereas matched MD native-contact values improved only the frozen model; Rosetta and ThermoMPNN scores gave no clear gain. Simulation data for experimental prediction should therefore be planned around controlled sequence changes and a quantity closely related to the measured property, rather than around data volume alone. The next test is to extend matched scans to more structures and evaluate the resulting models on new Tm measurements.

In the longer term, the same idea could guide an iterative design study. A model could choose which variant to calculate next, receive an FEP or MD result, and use that result to choose the following variant. Reinforcement learning is one way to make these sequential choices, but a computed score alone would not be a reliable reward. Our ESM2-proposed variants did not outperform random variants when both were scored by Rosetta, showing that a generator and a physics-based score do not by themselves make a useful design method. Any future reinforcement- learning study should therefore be compared with random sampling and simpler active-learning methods, and its success should be judged prospectively by new experimental Tm measurements.

Conflict of interest

T.M. is an employee of Epsilon Molecular Engineering, Inc. Y.M. is a scientific advisor of Epsilon Molecular Engineering, Inc.

Author contributions

Y.M. conceived and designed the study. Y.M., T.M., and K.S. carried out the overall computational study. Y.M. and T.M. analyzed the data. S.O. and K.O. performed the FEP calculations. Y.M. and T.M. wrote the manuscript. All authors discussed the results, reviewed the manuscript, and approved the final version.

Data availability

The source code for data processing, model training, analysis, and figure generation is available at `https://github.com/ymatsunaga/sim2real-nanobody-tm`. The processed data used in this study, together with reduced MD trajectories sufficient to reproduce the reported labels, will be available from Zenodo (DOI: `10.5281/zenodo.XXXXXXX`). Full raw simulation files are not included because of their size and are available from the corresponding author upon reasonable request.

Acknowledgements

This work was supported by JSPS KAKENHI Grant 23H03412 and JST NBDC Grant JPMJND2603, and partly supported by MEXT as "Program for Promoting Researches on the Supercomputer Fugaku" (Development and application of large-scale simulation-based inferences for biomolecules JPMXP1020230119). Computational resources were provided by the supercomputer Fugaku at the RIKEN Center for Computational Science under Project IDs hp230209, hp240215, and hp250233.

References

[1] Yamada, H., Liu, C., Wu, S., Koyama, Y., Ju, S., Shiomi, J., et al. Predicting materials properties with little data using shotgun transfer learning. ACS Central Science 5, 1717–1730 (2019). https://doi.org/10.1021/acscentsci.9b00804

[2] Aoki, Y., Wu, S., Tsurimoto, T., Hayashi, Y., Minami, S., Okubo, T., et al. Multitask machine learning to predict polymer–solvent miscibility using Flory–Huggins interaction parameters. Macromolecules 56, 5446–5456 (2023). https://doi.org/10.1021/acs.macromol.2c02600

[3] Minami, S., Hayashi, Y., Wu, S., Fukumizu, K., Sugisawa, H., Ishii, M., et al. Scaling law of Sim2Real transfer learning in expanding computational materials databases for real-world predictions. npj Computational Materials 11, 146 (2025). https://doi.org/10.1038/s41524-025-01606-5

[4] Muyldermans, S. Nanobodies: Natural single-domain antibodies. Annual Review of Biochemistry 82, 775–797 (2013). https://doi.org/10.1146/annurev-biochem-063011-092449
[5] Alexander, E. Leong, K. W. Discovery of nanobodies: a comprehensive review of their applications and potential over the past five years. Journal of Nanobiotechnology 22, 661 (2024). https://doi.org/10.1186/s12951-024-02900-y
[6] Kunz, P., Zinner, K., Mücke, N., Bartoschik, T., Muyldermans, S., Hoheisel, J. D. The structural basis of nanobody unfolding reversibility and thermoresistance. Sci. Rep. 8, 7934 (2018).
[7] Mezzasalma, T. M., Kranz, J. K., Chan, W., Struble, G. T., Schalk-Hihi, C., Deckman, I. C., et al. Enhancing recombinant protein quality and yield by protein stability profiling. J. Biomol. Screen. 12, 418–428 (2007).
[8] Kumar, S., Tsai, C. J., Nussinov, R. Factors enhancing protein thermostability. Protein Eng. 13, 179–191 (2000).
[9] Bekker, G.-J., Ma, B., Kamiya, N. Thermal stability of single-domain antibodies estimated by molecular dynamics simulations. Protein Science 28, 429–438 (2019). https://doi.org/10.1002/pro.3546
[10] Higashida, R. Matsunaga, Y. Enhanced conformational sampling of nanobody CDR H3 loop by generalized replica-exchange with solute tempering. Life (Basel) 11, 1428 (2021).
[11] Mohseni, A., Molakarimi, M., Taghdir, M., Sajedi, R. H., Hasannia, S. Exploring single-domain antibody thermostability by molecular dynamics simulation. J. Biomol. Struct. Dyn. 37, 3686–3696 (2019).
[12] Tomimoto, Y., Yamazaki, R., Shirai, H. Increasing the melting temperature of VHH with the in silico free energy score. Sci. Rep. 13, 4922 (2023).
[13] Rives, A., Meier, J., Sercu, T., Goyal, S., Lin, Z., Liu, J., et al. Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. Proceedings of the National Academy of Sciences of the United States of America 118, e2016239118 (2021). https://doi.org/10.1073/pnas.2016239118
[14] Lin, Z., Akin, H., Rao, R., Hie, B., Zhu, Z., Lu, W., et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. Science 379, 1123–1130 (2023). https://doi.org/10.1126/science.ade2574
[15] Hayes, T., Rao, R., Akin, H., Sofroniew, N. J., Oktay, D., Lin, Z., et al. Simulating 500 million years of evolution with a language model. Science 387, 850–858 (2025).
[16] Blaabjerg, L. M., Kassem, M. M., Good, L. L., Jonsson, N., Cagiada, M., Johansson, K. E., et al. Rapid protein stability prediction using deep learning representations. eLife 12, e82593 (2023). https://doi.org/10.7554/eLife.82593
[17] Li, M., Wang, H., Yang, Z., Zhang, L., Zhu, Y. DeepTM: A deep learning algorithm for prediction of melting temperature of thermophilic proteins directly from sequences. Computational and Structural Biotechnology Journal 21, 5544–5560 (2023). https://doi.org/10.1016/j.csbj.2023.11.006
[18] Alvarez, J. A. E. Dean, S. N. TEMPRO: nanobody melting temperature estimation model using protein embeddings. Scientific Reports 14, 19074 (2024). https://doi.org/10.1038/s41598-024-70101-6
[19] Savojardo, C., Manfredi, M., Martelli, P. L., Casadio, R. DDGemb: predicting protein stability change upon single- and multi-point variations with embeddings and deep learning. Bioinformatics 41 (2024).
[20] Ramon, A., Ni, M., Predeina, O., Gaffey, R., Kunz, P., Onuoha, S., et al. Prediction of protein biophysical traits from limited data: a case study on nanobody thermostability through NanoMelt. MAbs 17, 2442750 (2025).
[21] Mao, J., Song, Y., Kong, M., Guo, Y., Liu, Y., Pu, X. Thermostability prediction powered by synergistic deep learning at experimental and theoretical levels for nanobodies. ACS Appl. Mater. Interfaces 18, 6433–6451 (2026).
[22] Tsuboyama, K., Dauparas, J., Chen, J., Laine, E., Mohseni Behbahani, Y., Weinstein, J. J., et al. Mega-scale experimental analysis of protein folding stability in biology and design. Nature 620, 434–444 (2023). https://doi.org/10.1038/s41586-023-06328-6
[23] Chu, S. K. S., Narang, K., Siegel, J. B. Protein stability prediction by fine-tuning a protein language model on a mega-scale dataset. PLOS Computational Biology 20, e1012248 (2024). https://doi.org/10.1371/journal.pcbi.1012248
[24] Valdés-Tresanco, M. S., Valdés-Tresanco, M. E., Molina-Abad, E., Moreno, E. NbThermo: a new thermostability database for nanobodies. Database 2023, baad021 (2023). https://doi.org/10.1093/database/baad021
[25] Zhang, Y. Tsuda, K., NbBench: Benchmarking language models for comprehensive nanobody tasks, , (2025).
[26] Schmirler, R., Heinzinger, M., Rost, B. Fine-tuning protein language models boosts predictions across diverse tasks. Nature Communications 15, 7407 (2024). https://doi.org/10.1038/s41467-024-51844-2
[27] Phillips, J. C., Hardy, D. J., Maia, J. D. C., Stone, J. E., Ribeiro, J. V., Bernardi, R. C., et al. Scalable molecular dynamics on CPU and GPU architectures with NAMD. The Journal of Chemical Physics 153, 044130 (2020). https://doi.org/10.1063/5.0014475
[28] Best, R. B., Zhu, X., Shim, J., Lopes, P. E. M., Mittal, J., Feig, M., et al. Optimization of the additive CHARMM all-atom protein force field targeting improved sampling of the backbone ϕ, ψ and side-chain χ_1 and χ_2 dihedral angles. Journal of Chemical Theory and Computation 8, 3257–3273 (2012). https://doi.org/10.1021/ct300400x
[29] Jorgensen, W. L., Chandrasekhar, J., Madura, J. D., Impey, R. W., Klein, M. L. Comparison of sim-

ple potential functions for simulating liquid water. The Journal of Chemical Physics 79, 926–935 (1983). https://doi.org/10.1063/1.445869

[30] Humphrey, W., Dalke, A., Schulten, K. VMD: Visual molecular dynamics. Journal of Molecular Graphics 14, 33–38 (1996). https://doi.org/10.1016/0263-7855(96)00018-5

[31] Ramadoss, V., Dehez, F., Chipot, C. AlaScan: A graphical user interface for alanine scanning free-energy calculations. Journal of Chemical Information and Modeling 56, 1122–1126 (2016). https://doi.org/10.1021/acs.jcim.6b00162

[32] Essmann, U., Perera, L., Berkowitz, M. L., Darden, T., Lee, H., Pedersen, L. G. A smooth particle mesh Ewald method. The Journal of Chemical Physics 103, 8577–8593 (1995). https://doi.org/10.1063/1.470117

[33] Ryckaert, J.-P., Ciccotti, G., Berendsen, H. J. C. Numerical integration of the cartesian equations of motion of a system with constraints: molecular dynamics of n-alkanes. Journal of Computational Physics 23, 327–341 (1977). https://doi.org/10.1016/0021-9991(77)90098-5

[34] Best, R. B., Hummer, G., Eaton, W. A. Native contacts determine protein folding mechanisms in atomistic simulations. Proceedings of the National Academy of Sciences of the United States of America 110, 17874–17879 (2013). https://doi.org/10.1073/pnas.1311599110

[35] Tian, C., Kasavajhala, K., Belfon, K. A. A., Raguette, L., Huang, H., Migues, A. N., et al. ff19SB: amino-acid-specific protein backbone parameters trained against quantum mechanics energy surfaces in solution. Journal of Chemical Theory and Computation 16, 528–552 (2020). https://doi.org/10.1021/acs.jctc.9b00591

[36] Izadi, S., Anandakrishnan, R., Onufriev, A. V. Building water models: a different approach. The Journal of Physical Chemistry Letters 5, 3863–3871 (2014). https://doi.org/10.1021/jz501780a

[37] Eastman, P., Swails, J., Chodera, J. D., McGibbon, R. T., Zhao, Y., Beauchamp, K. A., et al. OpenMM 7: rapid development of high performance algorithms for molecular dynamics. PLOS Computational Biology 13, e1005659 (2017). https://doi.org/10.1371/journal.pcbi.1005659

[38] Dunbar, J., Krawczyk, K., Leem, J., Baker, T., Fuchs, A., Georges, G., et al. SAbDab: the structural antibody database. Nucleic Acids Research 42, D1140–D1146 (2014). https://doi.org/10.1093/nar/gkt1043

[39] Dolinsky, T. J., Nielsen, J. E., McCammon, J. A., Baker, N. A. PDB2PQR: an automated pipeline for the setup of Poisson–Boltzmann electrostatics calculations. Nucleic Acids Research 32, W665–W667 (2004). https://doi.org/10.1093/nar/gkh381

[40] Olsson, M. H. M., Søndergaard, C. R., Rostkowski, M., Jensen, J. H. PROPKA3: consistent treatment of internal and surface residues in empirical pK_a predictions. Journal of Chemical Theory and Computation 7, 525–537 (2011). https://doi.org/10.1021/ct100578z

[41] Kellogg, E. H., Leaver-Fay, A., Baker, D. Role of conformational sampling in computing mutation-induced changes in protein structure and stability. Proteins: Structure, Function, and Bioinformatics 79, 830–838 (2011). https://doi.org/10.1002/prot.22921

[42] Alford, R. F., Leaver-Fay, A., Jeliazkov, J. R., O'Meara, M. J., DiMaio, F. P., Park, H., et al. The Rosetta all-atom energy function for macromolecular modeling and design. Journal of Chemical Theory and Computation 13, 3031–3048 (2017). https://doi.org/10.1021/acs.jctc.7b00125

[43] Dieckhaus, H., Brocidiacono, M., Randolph, N. Z., Kuhlman, B. Transfer learning to leverage larger datasets for improved prediction of protein stability changes. Proceedings of the National Academy of Sciences of the United States of America 121, e2314853121 (2024). https://doi.org/10.1073/pnas.2314853121

[44] Kendall, A., Gal, Y., Cipolla, R. Multi-task learning using uncertainty to weigh losses for scene geometry and semantics. in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition 7482–7491, IEEE/CVF (2018).

[45] Murakami, T., Hashidate, Y., Matsunaga, Y. Mechanistic Interpretability of Fine-Tuned Protein Language Models for Nanobody Thermostability Prediction. bioRxiv (2025). Preprint. https://www.biorxiv.org/content/10.64898/2025.12.22.695878v1